

unit $\rightarrow 12, 0, 1, 2, 8, 10, 16, 18, 24, 26$ $\rightarrow$ K map for 5 variables

• Make two 4 variable K-maps

• If variables are A, B, C, D, E

BC \ DE	$\overline{D}\overline{E}$	$\overline{D}E$	DE	$D\overline{E}$
$\overline{B}\overline{C}$	0	1	3	2
$\overline{B}C$	4	5	7	6
BC	12	13	15	14
$B\overline{C}$	8	9	11	10

BC \ DE	$\overline{D}\overline{E}$	$\overline{D}E$	DE	$D\overline{E}$
$\overline{B}\overline{C}$	16	17	19	18
$\overline{B}C$	20	21	23	22
BC	28	29	31	30
$B\overline{C}$	24	25	27	26

$$A = \sum m(0, 2, 8, 10, 16, 18, 24, 26)$$

$$A = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$$

• overlapping concept use hoga

like if 0 & 16 dono p (1) hai to wo ek group
 bana skte ek map ko dusre ke upar rakhe
 aur overlap hojaye to group possible hai

$$Q \quad F(A, B, C, D, E) = \sum m(0, 2, 8, 10, 16, 18, 24, 26)$$

BC \ DE	$\overline{D}\overline{E}$	$\overline{D}E$	DE	$D\overline{E}$
$\overline{B}\overline{C}$	1			1
$\overline{B}C$				
BC				
$B\overline{C}$	1			1

$A = 0$

BC \ DE	$\overline{D}\overline{E}$	$\overline{D}E$	DE	$D\overline{E}$
$\overline{B}\overline{C}$	1			1
$\overline{B}C$				
BC				
$B\overline{C}$	1			1

$A = 1$

single group of 8

$$\text{Answer } A \rightarrow \overline{C}\overline{E} \cdot \sum m(0, 2, 8, 10, 16, 18, 24, 26)$$

Var B & C always on rows, D & E pathways on
 columns otherwise cell numbering changes
 & we get wrong answer

Q $F(A, B, C, D, E) = \sum m(0, 5, 6, 8, 9, 10, 11, 16, 20, 24, 25, 26, 27, 29, 31)$

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	0	1	3	2
$\bar{B}C$	4	5	7	6
BC	12	13	15	14
$B\bar{C}$	8	9	11	10

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	16	17	19	18
$\bar{B}C$	20	21	23	22
BC	28	29	31	30
$B\bar{C}$	24	25	27	26

Group 1 = $D\bar{A}C$

Group 2 = $D\bar{A}C$

Group 3 = $D\bar{A}C$

Group 4 = $D\bar{A}C$

Group 5 = $D\bar{A}C$

Group 6 = $D\bar{A}C$

Group 7 = $D\bar{A}C$

Group 8 = $D\bar{A}C$

Group 9 = $D\bar{A}C$

Group 10 = $D\bar{A}C$

Group 11 = $D\bar{A}C$

Group 12 = $D\bar{A}C$

Group 13 = $D\bar{A}C$

Group 14 = $D\bar{A}C$

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Group 96 = $D\bar{A}C$

Group 97 = $D\bar{A}C$

Group 98 = $D\bar{A}C$

Group 99 = $D\bar{A}C$

Group 100 = $D\bar{A}C$

lp-1 = $m(8, 9, 10, 11, 24, 25, 26, 27) = \bar{B}\bar{C}$

lp-2 = $m(25, 27, 29, 31) = ABE$

lp-3 = $m(0, 8, 16, 24) = \bar{D}\bar{E}\bar{C}$

lp-4 = $m(5) = \bar{A}\bar{B}C\bar{D}\bar{E}$

lp-5 = $m(16) = \bar{A}\bar{D}E\bar{B}C$

lp-6 = $m(20) = A\bar{B}\bar{E}\bar{D}$

$\Rightarrow F = \bar{B}\bar{C} + ABE + \bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}E + A\bar{B}\bar{E}\bar{D}$

Q $F(A, B, C, D, E) = \sum m(0, 1, 4, 5, 16, 17, 21, 25, 29)$

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	0	1	3	2
$\bar{B}C$	4	5	7	6
BC	12	13	15	14
$B\bar{C}$	8	9	11	10

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	16	17	19	18
$\bar{B}C$	20	21	23	22
BC	28	29	31	30
$B\bar{C}$	24	25	27	26

lp-1 = $(17, 21, 29, 25) = A\bar{D}E$

lp-2 = $(0, 1, 4, 5) = \bar{A}\bar{B}\bar{D}$

lp-3 = $(0, 1, 16, 17) = \bar{B}\bar{C}\bar{D}\bar{E}$

$F = \bar{A}\bar{B}\bar{D} + A\bar{D}E + \bar{B}\bar{C}\bar{D}\bar{E}$

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)

$$Q \quad F(A, B, C, D, E) = \sum m(0, 1, 2, 4, 6, 8, 10, 12, 13, 14, 17, 18, 20, 21, 22, 25, 26, 28, 29, 30) + d(5, 9, 15, 23, 27, 31)$$

	$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$		$\bar{D}\bar{E}$	$\bar{D}E$	DE	$D\bar{E}$
$\bar{B}\bar{C}$	1	0	X	1	3	10	1	2	8
$\bar{B}C$	1	1	X	1	5	6	1	22	20
BC	1	12	13	X	15	14	1	30	28
$B\bar{C}$	1	8	X	9	11	1	10	26	24

$\text{yp-1 } (0, 4, 12, 8, 16, 5, 13, 9) = \bar{A}\bar{D}$
 $\text{yp-2 } (1, 5, 13, 9, 17, 21, 29, 25) = \bar{D}E$
 $\text{yp-3 } (2, 6, 14, 10, 18, 22, 30, 26) = D\bar{E}$
 $\text{yp-4 } (20, 24, 22, 23, 28, 29, 30, 31) = \bar{B}AC$

$$F(A, B, C, D, E) = \bar{A}\bar{D} + \bar{D}E + D\bar{E} + \bar{B}AC$$

→ 6 variable 32K-maps

- Make four 4 variable K maps

	$\bar{E}\bar{F}$	$\bar{E}F$	EF	$E\bar{F}$
$\bar{C}\bar{D}$	0	1	3	2
$\bar{C}D$	4	5	7	6
CD	12	13	15	14
$C\bar{D}$	8	9	11	10

 $\bar{A}\bar{B}$

	$\bar{E}\bar{F}$	$\bar{E}F$	EF	$E\bar{F}$
$\bar{C}\bar{D}$	16	17	19	18
$\bar{C}D$	20	21	23	22
CD	28	29	31	30
$C\bar{D}$	24	25	27	26

 $\bar{A}B$

	$\bar{E}\bar{F}$	$\bar{E}F$	EF	$E\bar{F}$
$\bar{C}\bar{D}$	32	33	35	34
$\bar{C}D$	36	37	39	38
CD	44	45	47	46
$C\bar{D}$	40	41	43	42

 $\bar{A}\bar{B}$

	$\bar{E}\bar{F}$	$\bar{E}F$	EF	$E\bar{F}$
$\bar{C}\bar{D}$	48	49	51	50
$\bar{C}D$	52	53	55	54
CD	60	61	63	62
$C\bar{D}$	56	57	59	58

 $\bar{A}B$

d(11, 15, 27, 31, 41, 45)

Q F(ABCDEF) = m(1, 3, 5, 7, 9, 13, 17, 21, 25, 29, 33, 37, 49, 53, 57, 61)

	$\bar{E}\bar{F}$	$\bar{E}F$	$E\bar{F}$	EF
$\bar{C}\bar{D}$	0	1	3	2
$\bar{C}D$	4	5	7	6
CD	12	13	X	14
$C\bar{D}$	8	X	9	10

 $X \bar{A} \bar{B}$ $X \bar{A} \bar{B}$

	$\bar{E}\bar{F}$	$\bar{E}F$	$E\bar{F}$	EF
$\bar{C}\bar{D}$	16	17	19	18
$\bar{C}D$	20	21	23	22
CD	28	29	X	30
$C\bar{D}$	24	25	X	26

 $X \bar{A} \bar{B}$ $X \bar{A} \bar{B}$

	$\bar{E}\bar{F}$	$\bar{E}F$	$E\bar{F}$	EF
$\bar{C}\bar{D}$	32	33	35	34
$\bar{C}D$	36	37	39	38
CD	44	X	45	46
$C\bar{D}$	40	X	41	42

 $X \bar{A} \bar{B}$ $X \bar{A} \bar{B}$

	$\bar{E}\bar{F}$	$\bar{E}F$	$E\bar{F}$	EF
$\bar{C}\bar{D}$	48	49	51	50
$\bar{C}D$	52	53	55	54
CD	60	61	63	62
$C\bar{D}$	56	57	59	58

 $X \bar{A} \bar{B}$ $X \bar{A} \bar{B}$

gyp-1 (1, 5, 9, 13, 17, 21, 25, 29, 33, 37, 41, 45, 49, 53, 57, 61)

 $= \bar{E}\bar{F}$ gyp-2 (1, 5, 13, 9, 3, 7, 15, 11) = $\bar{A}\bar{B}F$ $F = \bar{A}\bar{B}F + \bar{E}\bar{F}$

→ Variable Entered Mapping (VEM)

- capable to represent higher variable functions with less no. of variable maps

	A	B	C	D	F
F=0 ←	0	0	0	0	0
F=1 ←	0	0	1	0	1
F=D ←	0	1	0	0	0
F=D ←	0	1	1	0	0
F=0 ←	1	0	0	0	0
F=X ←	1	0	1	0	X
F= $\bar{D}$ ←	1	1	0	0	1
F= $\bar{D}$ ←	1	1	1	0	1

⇒

	A	B	C	D	F
F=0 ←	0	0	0	0	0
F=1 ←	0	0	1	1	1
F=D ←	0	1	0	0	0
F=D ←	0	1	1	0	0
F=0 ←	1	0	0	0	0
F=X ←	1	0	1	0	X
F= $\bar{D}$ ←	1	1	0	0	1
F= $\bar{D}$ ←	1	1	1	0	1

	BC	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$
$\bar{A}$	0 ₀	1 ₁	D ₃	D ₂
A	0 ₄	X ₅	$\bar{D}$ ₇	$\bar{D}$ ₆

→ Steps to solve VEM

1. Take all variables in cell as '0' and get

SOP

0	1	0	0
0	X	0	0

$$= \bar{B}C$$

2. Make one variable of cell as '1' with all previous 1's as don't cares & get SOP

0	X	1	1
0	X	0	0

$$= \bar{A}B$$

3. Multiply SOP with variable

$$\Rightarrow \bar{A}B \cdot D$$

4. Repeat 2 & 3 until all variables are covered

0	X	0	0
0	X	1	1

$$= AB\bar{D}$$

5. SOP is sum of all SOP's

$$SOP = \bar{B}C + \bar{A}BDA + AB\bar{D}$$

X	0	X	0
1	1	X	X

$$SOP = \bar{A}B\bar{D} + \bar{A}BDA + AB\bar{D} = \bar{A}B$$

$$Q \quad f(A, B, C, D) = \sum m(2, 4, 3, 5, 13, 15) + d(8, 9, 10, 11)$$

	A	B	C	D	F	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{D}$	$\bar{A}$
--	---	---	---	---	---	-----------	-----------	-----------	-----------	-----------

0	0	0	0	0	0	1	1	1	1	1
---	---	---	---	---	---	---	---	---	---	---

1	0	0	0	1	0	1	1	1	0	1
---	---	---	---	---	---	---	---	---	---	---

2	0	0	1	0	1	1	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---

3	0	0	1	1	0	1	1	0	0	1
---	---	---	---	---	---	---	---	---	---	---

4	0	1	0	0	1	1	0	1	1	0
---	---	---	---	---	---	---	---	---	---	---

5	0	1	0	1	1	1	0	0	0	1
---	---	---	---	---	---	---	---	---	---	---

6	0	1	1	0	0	1	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---

7	0	1	1	1	0	1	0	0	0	1
---	---	---	---	---	---	---	---	---	---	---

8	1	0	0	0	X	0	1	1	1	X
---	---	---	---	---	---	---	---	---	---	---

9	1	0	0	1	X	0	1	1	0	X
---	---	---	---	---	---	---	---	---	---	---

10	1	0	1	0	X	0	1	0	1	X
----	---	---	---	---	---	---	---	---	---	---

11	1	0	1	1	X	0	1	0	0	X
----	---	---	---	---	---	---	---	---	---	---

12	1	1	0	0	0	0	0	1	1	0
----	---	---	---	---	---	---	---	---	---	---

13	1	1	0	1	1	0	0	0	0	1
----	---	---	---	---	---	---	---	---	---	---

14	1	1	1	0	0	0	0	0	1	0
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15	1	1	1	1	1	0	0	0	0	0
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min $\bar{B}\bar{C}\bar{D}, \bar{B}\bar{C}D, B\bar{C}\bar{D}, B\bar{C}D$ are 4 prime implicants

$\bar{A}$	0	1	0	1
A	X	X	0	0

$\bar{A}B\bar{D}$

X	0	X	0
1	1	X	0

0	1	0	1
X	X	0	0

$$= \bar{A}\bar{B}C + \bar{A}B\bar{C}$$

0	X	0	X
X	X	1	1

$$\bar{A}B\bar{A} + A\bar{D}\bar{A} + \bar{C}\bar{B} = \bar{A}\bar{B}C + \bar{A}B\bar{C}$$

$$SOP = AD + \bar{A}\bar{B}C + \bar{A}B\bar{C}$$

Q Reduce $f(A, B, C, D) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + AB\bar{C}D$ using VEM

• Jinme D nhi hai unke $(D + \bar{D})$ se * krdo

$$f = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D}$$

	A	B	C	D	f
$\bar{A}\bar{B}\bar{C}\bar{D}$	0	0	0	0	1
$\bar{A}\bar{B}\bar{C}D$	0	0	0	1	1
$\bar{A}\bar{B}C\bar{D}$	0	0	1	0	1
$\bar{A}\bar{B}CD$	0	0	1	1	1
$\bar{A}B\bar{C}\bar{D}$	0	1	0	0	1
$\bar{A}B\bar{C}D$	0	1	0	1	1
$\bar{A}BC\bar{D}$	0	1	1	0	1
$\bar{A}BCD$	0	1	1	1	1
$A\bar{B}\bar{C}\bar{D}$	1	0	0	0	0
$A\bar{B}\bar{C}D$	1	0	0	1	0
$A\bar{B}C\bar{D}$	1	0	1	0	0
$A\bar{B}CD$	1	0	1	1	0
$AB\bar{C}\bar{D}$	1	1	0	0	0
$AB\bar{C}D$	1	1	0	1	0
$ABC\bar{D}$	1	1	1	0	0
$ABCD$	1	1	1	1	0

1	1	1	1
0	0	0	0

$$= \bar{A}\bar{B}\bar{C}$$

X	X	X	X
0	0	0	1

$$= B\bar{C} \cdot D$$

X	0	0	0
1	0	0	0

Q. Reduce $\bar{A}\bar{B}\bar{C}E + \bar{A}\bar{B}C + \bar{A}BCF + \bar{A}B\bar{C} + A\bar{B}CD + AB\bar{C}F$ using VEM method.

3 variables (A, B, C) appear in all terms so consider D, E, F as map entered variables.

	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	E	1	F	1
A	0	D	0	$\bar{F}$

0	1	0	1
0	0	0	0

$$\bar{A}\bar{B}C + \bar{A}B\bar{C}$$

0	X	0	X
0	1	0	0

$$= \bar{B}\bar{C} \cdot D$$

1	X	0	X
0	0	0	0

$$= \bar{B}\bar{A} \cdot E$$

0	X	1	X
0	0	0	0

$$= \bar{A}C \cdot F$$

0	X	0	X
0	0	0	1

$$= B\bar{C} \cdot \bar{F}$$

$$\Rightarrow f = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{B}\bar{C}D + \bar{B}\bar{A}E + \bar{A}CF + B\bar{C}\bar{F}$$

Q solve $F = m_0 + m_1 F + m_2 + m_4 E + m_6 (E + E') + m_7 F + m_{10} E + m_{12} + m_{15} F + d. (m_5 F + m_9 + m_{11} \bar{E} + m_8 E)$.

1) F must be 4 variables (A, B, C, D) func with map entered variables E & F .

$\bar{A}\bar{B}$	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1_0	F_1	3	1_2
$\bar{A}B$	E_4	$F_2 X_5$	F_7	$E + \bar{E}_6$
AB	(F_{12})	11_{13}	F_{15}	E_{14}
$A\bar{B}$	$E_2 X_8$	X_9	$\bar{E}_2 X_{11}$	E_{10}

1	F	0	1
E	$F_2 \phi$	F	1
1	0	F	0
$E_2 \phi$	ϕ	$\bar{E}_2 \phi$	E

1	0	0	1
0	0	1	1
1	0	0	0
0	ϕ	ϕ	0

$$\bar{A}C\bar{D} + \bar{A}\bar{B}C\bar{D} + AB\bar{C}\bar{D}$$

X	0	0	X
1	0	0	X
X	0	0	0
$E_2 X$	X	$\bar{E}_2 X$	1

X	1	0	X
0	$F_2 X$	1	X
X	0	1	0
0	X	0	0

$A B C D$	min	max
0000	0m	0
1000	0m	0
0001	2m	0
1100	2m	0
1001	2m	0
1110	2m	0
1101	2m	0

$$(\bar{C}\bar{D} + \bar{A}\bar{B}\bar{D})E$$

$$F = (BCD + \bar{A}\bar{B}\bar{C})F$$

→ Quine Mc cluskey minimisation technique
(Tabular method)

- Implicant : group of 1's
- Prime implicant : Largest possible group of 1's
- Essential Prime implicant : implicant having atleast one minterm that can't be combined in any other way

eg → $Y(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$

	A	B	C	D	
0	0	0	0	0	0 ones
1	0	0	0	1	1 one
3	0	0	1	1	2 ones
7	0	1	1	1	3 ones
8	1	0	0	0	
9	1	0	0	1	
11	1	0	1	1	
15	1	1	1	1	

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}CD + ABCD$$

Step-1 Make a table by arranging all min terms according to no of 1's contained

group	min term	A B C D
0	m_0	0 0 0 0
1	m_1	0 0 0 1
	m_8	1 0 0 0
2	m_3	0 0 1 1
	m_9	1 0 0 1
3	m_7	0 1 1 1
	m_{11}	1 0 1 1
4	m_{15}	1 1 1 1

put tick mark over matched pairs (step-2)

✓

✓

✓

✓

✓

✓

✓

✓

→ If matched put (-) on changed bit & tick in table - 1

• Step + 2 :-

→ If n^{th} & $(n+1)^{\text{th}}$ group differ by one bit

Binary Rep.

Group

Matched Pairs

A B C D

0

$m_0 - m_1$

0 0 0 - 1 ✓

$m_0 - m_8$

0 0 0 - 1 ✓

1

$m_1 - m_3$

0 0 - 1 ✓

$m_1 - m_9$

0 0 1 - 1 ✓

$m_8 - m_9$

1 0 0 - ✓

2

$m_3 - m_7$

0 - 1 1 ✓

$m_3 - m_{11}$

- 0 1 1 ✓

$m_9 - m_{11}$

1 0 - 1 ✓

3

$m_7 - m_{15}$

- 1 1 1 ✓

$m_{11} - m_{15}$

1 - 1 1 ✓

• Step → 3

Again check for matches amongst matched pairs & repeat the same until matching stops

Group

M.O.

Binary

A B C D

0

$m_0 - m_1 - m_8 - m_9$

- 0 0 -

$m_0 - m_8 - m_1 - m_9$

- 0 0 -

} $\bar{B} \bar{C}$

1

$m_1 - m_3 - m_9 - m_{11}$

- 0 - 1

$m_1 - m_9 - m_3 - m_{11}$

- 0 - 1

} $\bar{B} D$

2

$m_3 - m_7 - m_{11} - m_{15}$

- 1 1 1

$m_3 - m_{11} - m_7 - m_{15}$

- 1 1 1

} CD

No further matching can be done

After matching stops, all unmatched pairs from every step become prime implicants

• Step 24

Make prime implicant table to find the essential prime implicants

given min terms

P.I	Min terms involved	0	1	3	7	8	9	11	15
$\overline{B}\overline{C}$	0, 1, 8, 9	X	X			X	X		
$\overline{B}D$	1, 3, 9, 11		X	X				X	X
CD	3, 7, 11, 15			X	X			X	X

• Put 'x' where those minterms are used in that PI

Circle the 'x' which is alone in a column

• Find prime implicants ki rows me 'x' ayega woh essential prime implicant honge.

$$\Rightarrow Y = \overline{B}\overline{C} + CD$$

• If don't care terms hai to unhe bi min terms maanke matching kro but

don't use them in prime implicant table

Q $f = \sum m(6, 7, 8, 9) + d(-10, 11, 12, 13, 14, 15)$

$6 \rightarrow 0110 \rightarrow 2$ $21m - 11 \rightarrow 11011 \rightarrow 3$

$7 \rightarrow 0111 \rightarrow 3$ $21m - 12 \rightarrow 11100 \rightarrow 2$

$8 \rightarrow 11000 \rightarrow 1$ $21m - 13 \rightarrow 11101 \rightarrow 3$

$9 \rightarrow 1001 \rightarrow 2$ $14 \rightarrow 1110 \rightarrow 3$

$10 \rightarrow 1010 \rightarrow 2$ $15 \rightarrow 1111 \rightarrow 4$

group	minimum	ABCD	
0 1 1	$21m - m_8 - pm - 8m$	1 0 0 0	✓
0 2 1	$11m - m_6 - pm - 8m$	0 1 1 0	✓
0 1 1	$11m - m_9 - pm - 8m$	1 0 0 1	✓
0 1 1	$21m - m_{10} - pm - 8m$	1 0 1 0	✓
0 1 1	$11m - m_{12} - pm - 8m$	1 1 0 0	✓
3	m_7	0 1 1 1	✓
1 1 1	$21m - m_{11} - pm - 8m$	1 0 1 1	✓
1 1 1	$21m - m_{13} - pm - 8m$	1 1 0 1	✓
1 1 1	$21m - m_{14} - pm - 8m$	1 1 1 0	✓
4	$21m - m_{15} - pm - 8m$	1 1 1 1	✓
1 1 1	$21m - pm - 11m - 01m$		
1 1 1	$21m - pm - 11m - 01m$		

group	Matched Pairs	ABCD	
0	$21m - m_8 - m_9$	1 0 0 -	
	$\checkmark m_8 - m_{10}$	1 0 - 0	
	$\checkmark m_8 - m_{12}$	1 - 0 0	
2	$\checkmark m_6 - m_7$	0 1 1 -	
	$\checkmark m_6 - m_{14}$	- 1 1 0	
	$\checkmark m_9 - m_{11}$	1 0 - 1	
	$\checkmark m_9 - m_{13}$	1 - 0 1	
	$\checkmark m_{10} - m_{11}$	1 0 1 -	
	$\checkmark m_{10} - m_{14}$	1 - 1 0	
	$\checkmark m_{12} - m_{13}$	1 0 1 -	
	$\checkmark m_{12} - m_{14}$	1 1 - 0	

→ Number system

1. Decimal no system (10) 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
2. Binary (2) 1, 0
3. Octal (8) 0, 1, 2, 3, 4, 5, 6, 7
4. Hexadecimal (16) 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
(10) (11) (12) (13) (14) (15)

Q $(11101)_2 = (?)_{10}$

$$11101$$

$$= 1 \times 2^0 + 0 \times 2^1 + 1 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 = 0$$

$$= 1 + 0 + 4 + 8 + 16 = 29$$

$$= 29$$

Q $(10101.101)_2 = (?)_{10}$

$$[1 \times 2^0 + 0 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 1 \times 2^4] + 0.5 [1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}]$$

$$[1 + 4 + 16] + [0.5] = 21 + [5/8]$$

$$\frac{5}{8} = 0.625$$

$$= 21.625$$

Q $(32)_{10} = (?)_2$

$$\begin{array}{r|l} 32 & \\ \hline 0 & 16 \\ 0 & 8 \\ 0 & 4 \\ 0 & 2 \\ 0 & 1 \end{array}$$

÷ by 2 & write quotient
and remainder

$$(100000)_2$$

P. 3. 1 $26 + 0.407$

(01)

0	26
---	----

1/13

06

1	3
---	---

1

$0.407 \times 2 =$

$$0.814 \times 2$$

$$0.628 \times 2$$

$$0.256 \times 2$$

$$E = x \underline{0.814} \times 1 \rightarrow 10 \times 0$$

$$= 1.628$$

$$= 1.256$$

$$= 0.412$$

$$(01(9)) = (101 \cdot 10101)$$

Q $(\underline{11010001111})_2 = (?)_{16}$

- Make group of 4 starting from LSB
- Write its respective hexadecimal value

$1101 + 0001 + 1111 \text{ sys} = (D1F)_{16}$
 $\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $13 \rightarrow D \quad \quad \quad 1 \quad \quad \quad 15 \rightarrow F$

Q $\underbrace{0101}_5 \underbrace{1110}_E \underbrace{0010}_2 = (5E2)_{16}$

Q 01011 · 1010 101 · 10110100
 5 D 5 8 4

$$(5D5.B4)_{16}$$

Q $(319E)_{16} \rightarrow (?)_2$

$$3 \rightarrow 0011$$

$$9 \rightarrow 1001$$

$$1 \rightarrow 0001$$

$$E \rightarrow 1110 = 0 + 0$$

$$= (0011\ 0001\ 1001\ 1110)_2$$

$$1 = 1 + 0$$

$$1 = 0 + 1$$

Q $(9AE)_{16} = (?)_{10}$

$$9 = 1 + 1 + 1$$

$$11 = 1 + 1 + 1$$

$$= E \times 16^0 + A \times 16^1 + 9 \times 16^2$$

$$= 14 \times 16^0 + 10 \times 16^1 + 9 \times 16^2$$

$$= 14 + 160 + 2304$$

$$= (2478)_{10}$$

$$110011$$

$$0011$$

$$0111 +$$

$$00011$$

$$100011$$

$$01011$$

Q $(928)_{10} = (?)_{16}$

$$\begin{array}{r|l} 0 & 928 \\ 10 & 58 \\ & 3 \end{array}$$

$$= (3A0)_{16}$$

$$11$$

$$11$$

$$11$$

$$01 +$$

$$1001$$

Q $545.789 = (?)_{16}$

$$\begin{array}{r|l} 1 & 545 \\ 2 & 34 \\ & 2 \end{array}$$

$$545 = 221$$

$$0 = 0 - 0$$

$$1 = 0 - 1$$

$$0 = 1 - 1$$

$$1 = 1 - 01$$

$$11 = 1 - 0$$

$$0.789 \times 16 = 12.62401 \rightarrow 12 \rightarrow 0C$$

$$0.624 \times 16 = 9.98401 \rightarrow 9 \rightarrow 9$$

$$0.984 \times 16 = 15.74400 \rightarrow 15 \rightarrow 0F$$

$$0.744 \times 16 = 11.884 \rightarrow 11 \rightarrow B$$

$$(221.C9FB)_{16}$$

$$1110$$

$$1110$$

$$1010$$

$$1010$$

C9FB

→ Binary Arithmetic

1. Addition

$0 + 0 = 0$

$0 + 1 = 1$

$1 + 0 = 1$

$1 + 1 = 10$

$1 + 1 + 1 = 11$

$$\begin{array}{r} 1100 \\ + 1110 \\ \hline 11010 \end{array}$$

$$\begin{array}{r} 1100 \\ + 1110 \\ \hline 11010 \end{array}$$

$$\begin{array}{r} 11 \\ + 10 \\ \hline 101 \end{array}$$

$$\begin{array}{r} 11 \\ + 10 \\ \hline 101 \end{array}$$

$$\begin{array}{r} 11 \\ + 10 \\ \hline 101 \end{array}$$

2. Subtraction

$0 - 0 = 0$

$1 - 0 = 1$

$1 - 1 = 0$

$10 - 1 = 1$

$0 - 1 = 11$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 10 \\ 011 \\ 001010 \\ + 100 \\ \hline 0111 \end{array}$$

$$0101$$

$$\begin{array}{r} 1010 \\ 00010 \\ + 100 \\ \hline 111 \end{array}$$

$$0111$$

3. Multiplication

$$0 \times 0 = 0$$

$$1 \times 0 \text{ or } 0 \times 1 = 0$$

$$1 \times 1 = 1$$

$$1110 \leftarrow \Delta^1$$

$$10010 \leftarrow \Delta^2$$

$$\times 101$$

$$0100100$$

$$00000 \times \leftarrow \Delta^1$$

$$10100 \times \leftarrow \Delta^2$$

$$110010$$

$$1010 \leftarrow \Delta^1$$

$$1001 \leftarrow \Delta^2$$

$$\times 11$$

$$10011$$

$$100100 \times \leftarrow \Delta^1$$

$$111000 \leftarrow \Delta^2$$

4. Division

$$110 \overline{) 10101000111}$$

$$\begin{array}{r} 10 \\ -0 \\ \hline 101 \end{array}$$

$$\begin{array}{r} 101 \\ -0 \\ \hline 1010 \end{array}$$

$$\begin{array}{r} 1010 \\ -110 \\ \hline 1001 \end{array}$$

$$\begin{array}{r} 1001 \\ -110 \\ \hline 0010 \end{array}$$

$$\begin{array}{r} 0010 \\ -110 \\ \hline 000 \end{array}$$

$$000$$

$$100 =$$

→ 1's complement and 2's complement

• 1's complement
change 0 to 1 and 1 to 0

• 2's complement
Add 1 to 1's complement

Q 1010

1's → 0101

2's → 10101

0110001

1's → 0111

2's → 1000

Q 11110

1's → 00001

2's → 00011

0010010

1's → 11010

2's → 11101

→ Subtraction by 1's complement

1. Find binary of A
2. Find 1's complement of B
3. Add them
4. If carry 1 is present, answer is +ve (end around carry)
5. If carry is absent, ans is -ve (find 1's complement)

Q 12 - 9

A → 1100

B → 1001

$\bar{B} = B(1's) \rightarrow 0110$

$A + \bar{B} = 1100 + 0110$

01100101

carry ← 10010100

(+ve)

add carry to LSB

= 0011

Q 9 - 12
 9 $\rightarrow$ 1001 (A) A for given bit
 12 $\rightarrow$ 1100 (B) B for given bit
 ($\bar{B}$) = 0011
 + 1001
+ 0011
 1100
 carry absent
 Ans $\rightarrow$ -ve
 $\Rightarrow$ Ans = -0011

Q 111011
 - 101

0010 = 1 + 1100 $\rightarrow$ $\bar{B}$

A $\rightarrow$ 111011

B $\rightarrow$ 00101

$\bar{B} \rightarrow$ 11010

A + $\bar{B} \rightarrow$ 111011

111011

1110101

1110101

= 1110101

(ev -) 1100 =

Q 00011
 - 11111

A $\rightarrow$ 00011

B $\rightarrow$ 11111

$\bar{B} \rightarrow$ 00000

A + $\bar{B} =$ 00011

+ 00000

00011

-ve

No carry

1's comp = -11100

→ Subtraction using 2's complement

- Find binary of A (A) 1001 = 9
- Find 2's complement of B 0011 = 3
- Add them 1100 = (A)
- If carry 1 is present, ans is +ve, delete the carry 1100 +
- If carry is 0, ans is -ve, find 2's complement of the ans.

W - - - - - ans

Q 9 - 12

1100 - = ans. E

9 → 1001 A

12 → 1100 B

110111

101 -

$\bar{B} \rightarrow 0011 + 1 = 0100$

ans. sum of 10

110111 → A

1001010111 → B

101000 → B

+ 0100

110111 → $\bar{B} + A$

1101

0101 NO carry

✓ ans. 01

1010111 ans

2's comp → 00010 + 1 =
= 0011 (-ve)

11000000 → A

11000

11111 → B

11111

00000 → $\bar{B}$

110000

= $\bar{B} + A$

+ 000000 +

✓ ans. 01

110000

110000

001111 = -ve

→ codes in digital system

1. Binary coded decimal (BCD code)

- write decimal nos in 4 bits

1 → 0001

1000

← 11

2 → 0010

3 → 0011

4 → 0100

0010 1000

5 → 0101

6 → 0110

7 → 0111

8 → 1000

9 → 1001

(35)₁₀ → (0011 0101)_{BCD}

(256)₁₀ → (0010 0101 0110)_{BCD}

2. American standard code for information interchange (ASCII) code

- 7 bit code & represents 128 characters

Ascii code for

LUV is

1001100 01010101 1010110

↓ ↓ ↓

L U V

3. extended Binary coded decimal interchange code (EBCDIC code)

- 8 bit code showing 256 characters

→ Binary to BCD

- 35 → 0011 0101
- 17 → 0001 0111

→ BCD to Binary

- 0001 0100
 $\downarrow \quad \downarrow$
 1 4

= 14

- 01001001 = 49
 $\downarrow \quad \downarrow$
 4 9

⇒ XS-13 code
 Decimal $\xrightarrow{\text{BCD}}$ + 3 $\rightarrow$ XS-3 code
 (0011)

1 → 0100 (4)

2 → 0101 (5)

3 → 0110 (6)

4 → 0111 (7)

5 → 1000 (8)

6 → 1001 (9)

7 → 1010 (10)

8 → 1011 (11)

9 → 1100 (12)

0 → 1101 (13)

10 → 1110 (14)

11 → 1111 (15)

12 → 0000 (0)

13 → 0001 (1)

14 → 0010 (2)

15 → 0011 (3)

- Gray code (Reflected Binary code) -
- Only 1 bit difference in 2 successive values

Decimal	Binary	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000
16	10000	0000

→ Binary to Gray code conversion

1. Record MSB as it is
2. Add MSB to next bit, record sum & neglect carry
3. Repeat

1110 → 1001

1101 → 1011
MSB LSB

1011 → 1100

→ signed number system has two parts
 MSB → 0 → +ve no
 MSB → 1 → -ve no

01111	→ +15	decimal
00000	→ 15	0
01111	→ -15	1
10000		2
01000		3
01100		4
11100		5
01000		6
00100		7
00101		8
10101		9
11101		10
01101		11
01011		12
11011		13
10011		14
00011		15
00000		16

... has ... at ...
 ... is ...
 ... has ...
 ...
 ...
 ...

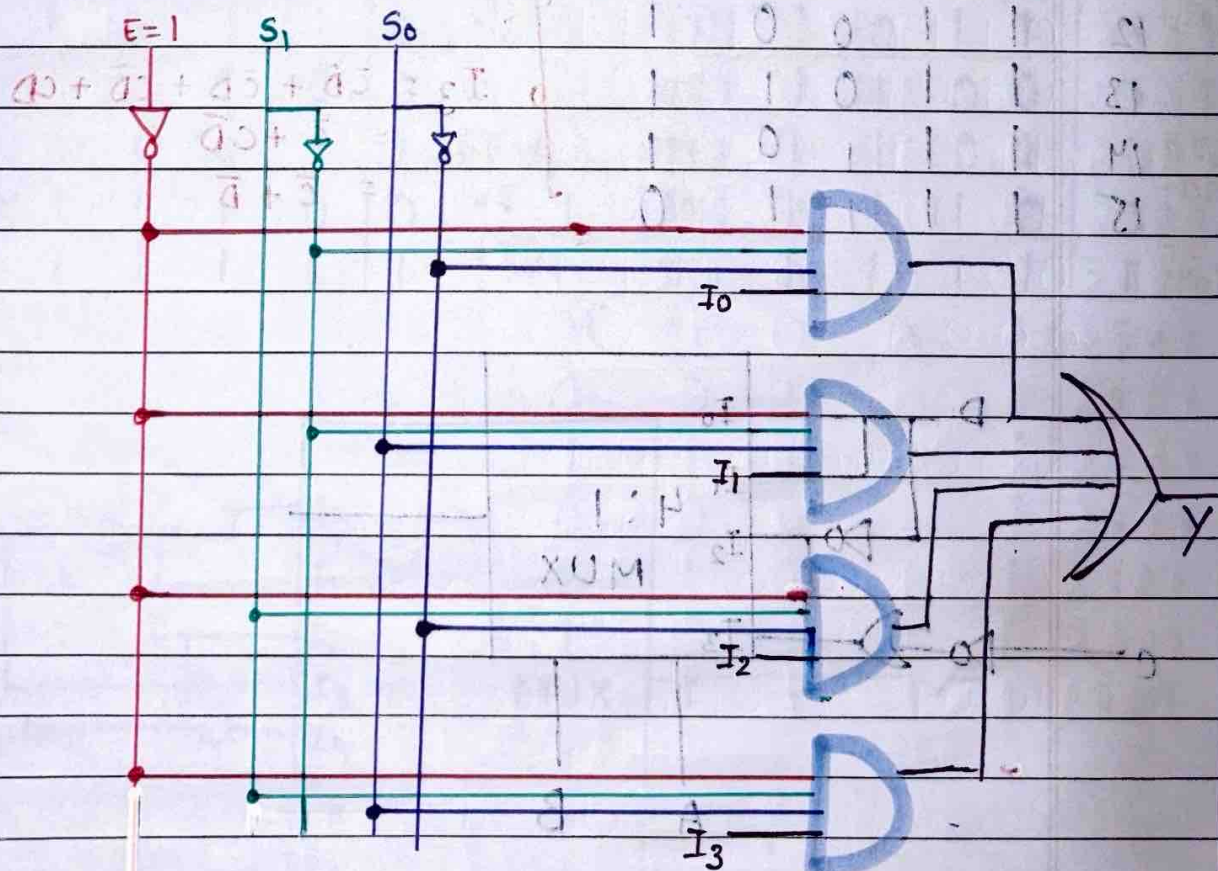
0101 ← 1011
 0011 ← 1101

→ Multiplexer (MUX) = (Data Selector)

- Multiplexer is a combinational circuit that takes multiple input signals to give 1 output signal
- Also called data selector
- For n select line, 2^n data inputs are present
- MUX only works when enable pin is active low ($E=0$)

E	S ₁	S ₀	Y
0	0	0	I ₀
0	0	1	I ₁
0	1	0	I ₂
0	1	1	I ₃
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

$$Y = \bar{E} (\bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3)$$

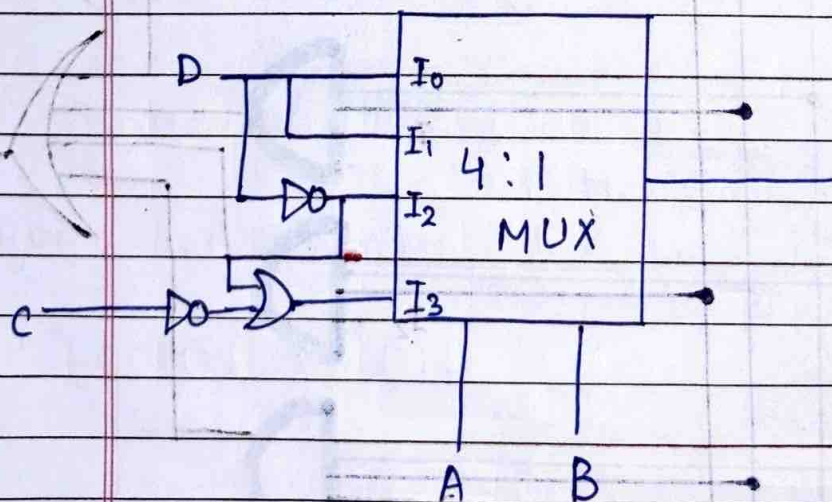


4:1 MUX

Q Implement $F(A, B, C, D) = \sum m(1, 3, 5, 7, 8, 10, 12, 13, 14)$ using 4:1 MUX.

a) select lines $S_1 = A$ and $S_0 = B$

	A	B	C	D	F	
0	0	0	0	0	0	$\bar{C}\bar{D} + \bar{C}D = \bar{C}$
1	0	0	0	1	0	$I_0 = \bar{C}$
2	0	0	1	0	0	
3	0	0	1	1	1	
4	0	1	0	0	0	$\bar{C}D + CD = D$
5	0	1	0	1	1	$I_1 = D$
6	0	1	1	0	0	
7	0	1	1	1	1	
8	1	0	0	0	1	$\bar{C}\bar{D} + C\bar{D} = \bar{D}$
9	1	0	0	1	0	$I_2 = \bar{D}$
10	1	0	1	0	1	
11	1	0	1	1	0	
12	1	1	0	0	1	
13	1	1	0	1	1	$I_3 = \bar{C}\bar{D} + \bar{C}D + C\bar{D} + CD$
14	1	1	1	0	1	$= \bar{C} + C\bar{D}$
15	1	1	1	1	0	$= \bar{C} + \bar{D}$

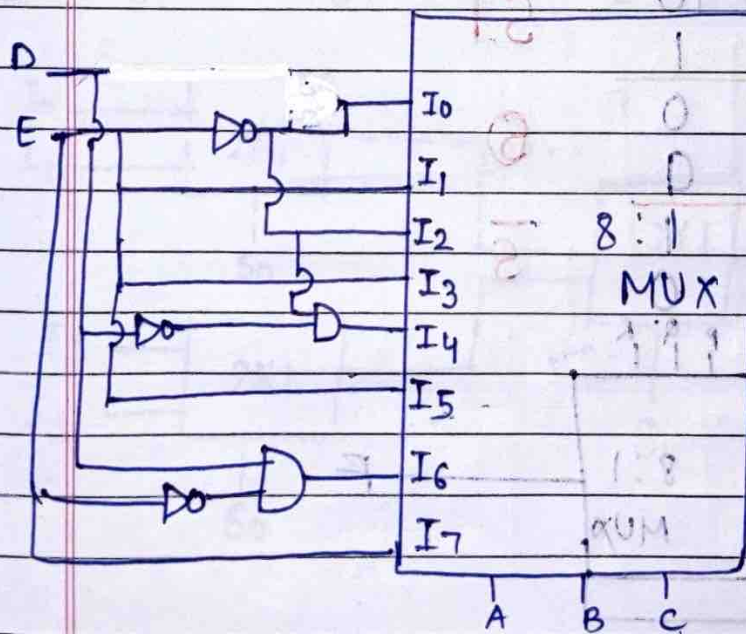


(A, B, C, D, E)

Q Implement $F = \sum m(0, 2, 5, 7, 8, 10, 13, 15, 16, 21, 23, 26, 29, 31)$ using 8:1 MUX choosing A, B, C as select lines.

using 8:1 MUX choosing A, B, C as select lines.

A	B	C	D	E	F	A	B	C	D	E	F
0	0	0	0	0	1	0	0	0	0	0	1
1	0	0	0	1	0	0	0	0	1	0	0
2	0	0	1	0	1	0	0	1	0	0	1
3	0	0	1	1	0	0	0	1	1	0	0
4	0	1	0	0	0	0	1	0	0	0	0
5	0	1	0	1	1	0	1	0	1	1	1
6	0	1	1	0	0	0	1	1	0	0	0
7	0	1	1	1	1	0	1	1	1	1	1
8	1	0	0	0	0	1	0	0	0	0	0
9	1	0	0	1	0	1	0	0	1	0	0
10	1	0	1	0	0	1	0	1	0	0	0
11	1	0	1	1	1	1	0	1	1	1	1
12	1	1	0	0	0	1	1	0	0	0	0
13	1	1	0	1	1	1	1	0	1	1	1
14	1	1	1	0	0	1	1	1	0	0	0
15	1	1	1	1	1	1	1	1	1	1	1



Q. Design $F(P, Q, R, S) = \bar{P}R + \bar{Q}S + PR\bar{S}$ using 8:1 MUX with select inputs PQR

$$\textcircled{1} \bar{P}R(Q + \bar{Q})(S + \bar{S}) = (\bar{P}RQ + \bar{P}R\bar{Q})(S + \bar{S})$$

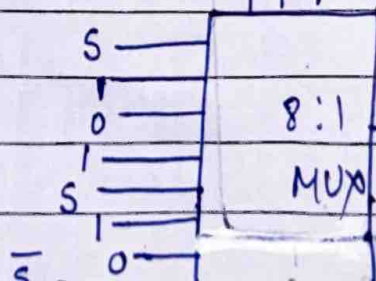
$$= \bar{P}RQS + \bar{P}RQ\bar{S} + \bar{P}\bar{R}QS + \bar{P}\bar{R}Q\bar{S}$$

$$\textcircled{2} \bar{Q}S(P + \bar{P})(R + \bar{R}) = (\bar{Q}SP + \bar{Q}S\bar{P})(R + \bar{R})$$

$$= \bar{Q}SPR + \bar{Q}S\bar{P}R + \bar{Q}SP\bar{R} + \bar{Q}S\bar{P}\bar{R}$$

$$\textcircled{3} PR\bar{S}(Q + \bar{Q}) = PR\bar{S}Q + PR\bar{S}\bar{Q}$$

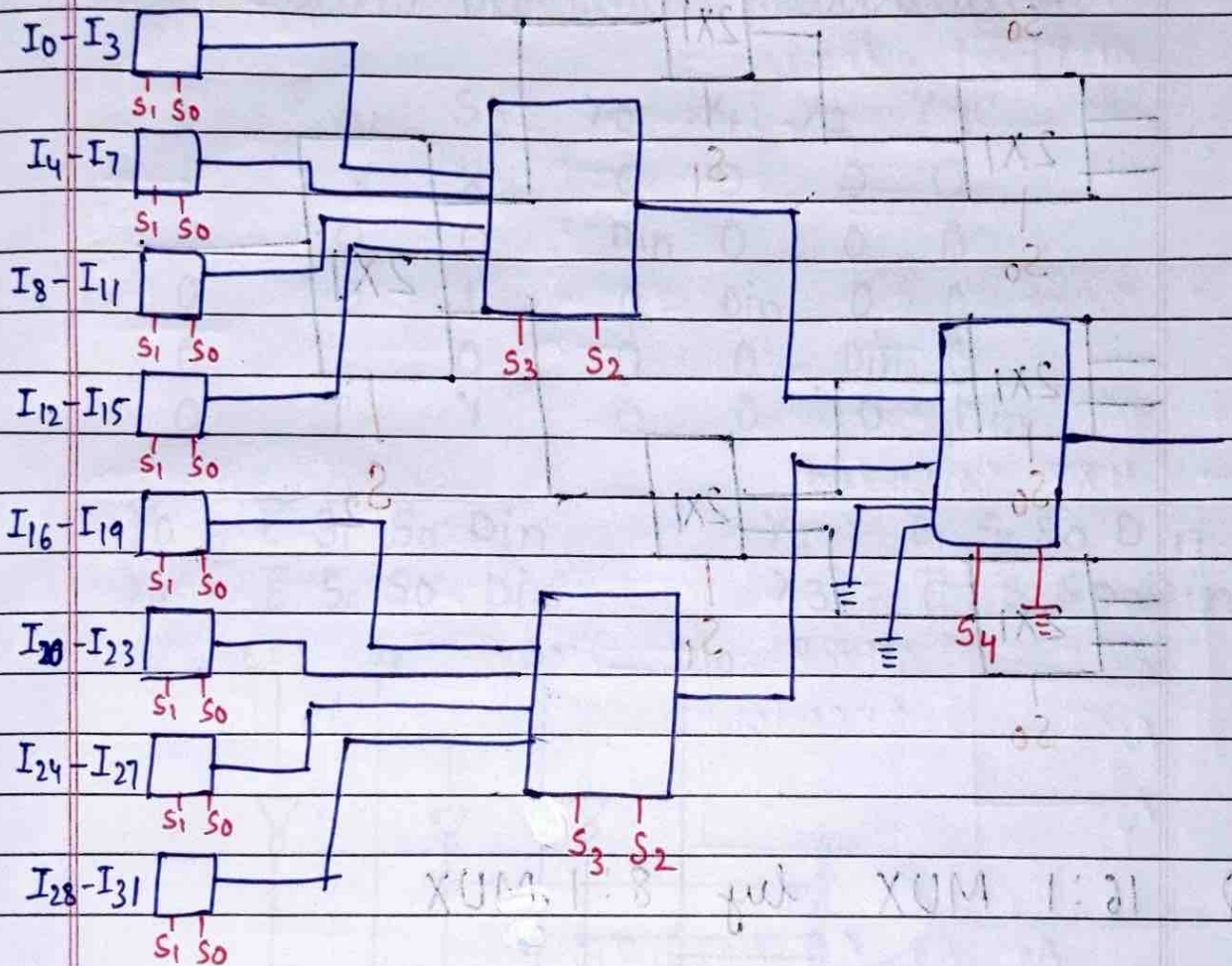
	P	Q	R	S	F
$\bar{P}\bar{Q}\bar{R}\bar{S}$	0	0	0	0	0
$\bar{P}\bar{Q}\bar{R}S$	0	0	0	1	1
$\bar{P}\bar{Q}R\bar{S}$	0	0	1	0	1
$\bar{P}\bar{Q}RS$	0	0	1	1	1
$\bar{P}Q\bar{R}\bar{S}$	0	1	0	0	0
$\bar{P}Q\bar{R}S$	0	1	0	1	0
$\bar{P}QR\bar{S}$	0	1	1	0	1
$\bar{P}QRS$	0	1	1	1	1
$P\bar{Q}\bar{R}\bar{S}$	1	0	0	0	0
$P\bar{Q}\bar{R}S$	1	0	0	1	1
$P\bar{Q}R\bar{S}$	1	0	1	0	1
$P\bar{Q}RS$	1	0	1	1	1
$PQ\bar{R}\bar{S}$	1	1	0	0	0
$PQ\bar{R}S$	1	1	0	1	0
$PQR\bar{S}$	1	1	1	0	1
$PQRS$	1	1	1	1	1



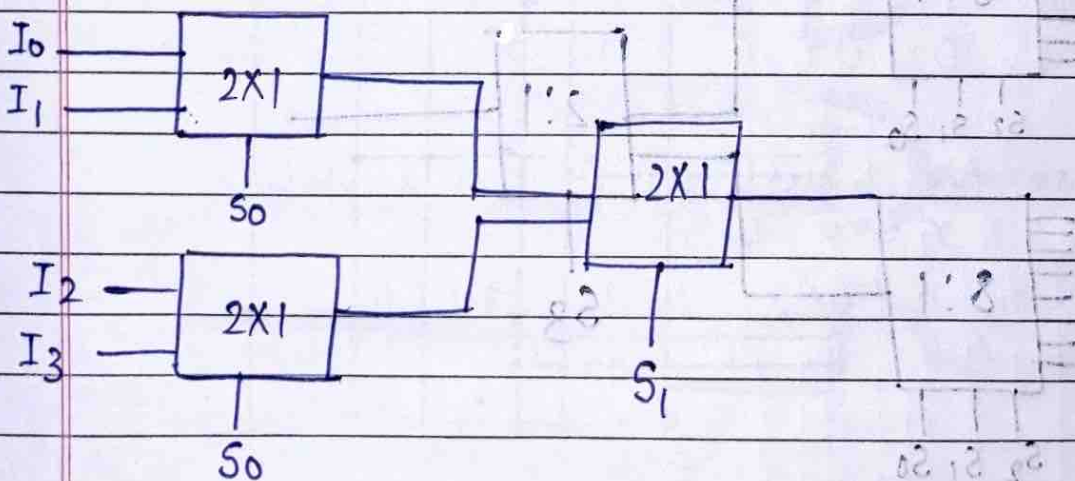
Q Design 32:1 MUX by 4:1 MUX only

↓
32 input
5 select lines

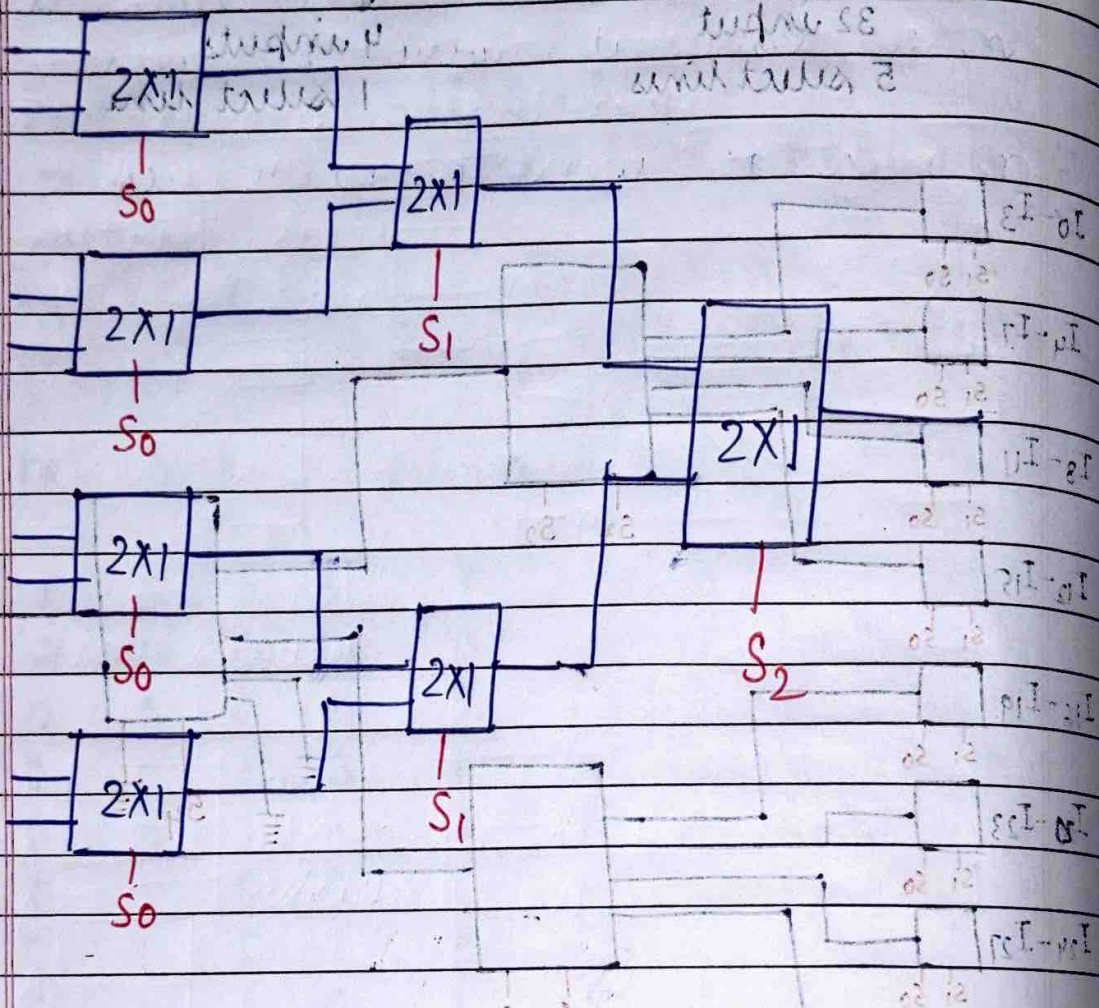
↓
4 inputs
1 select line



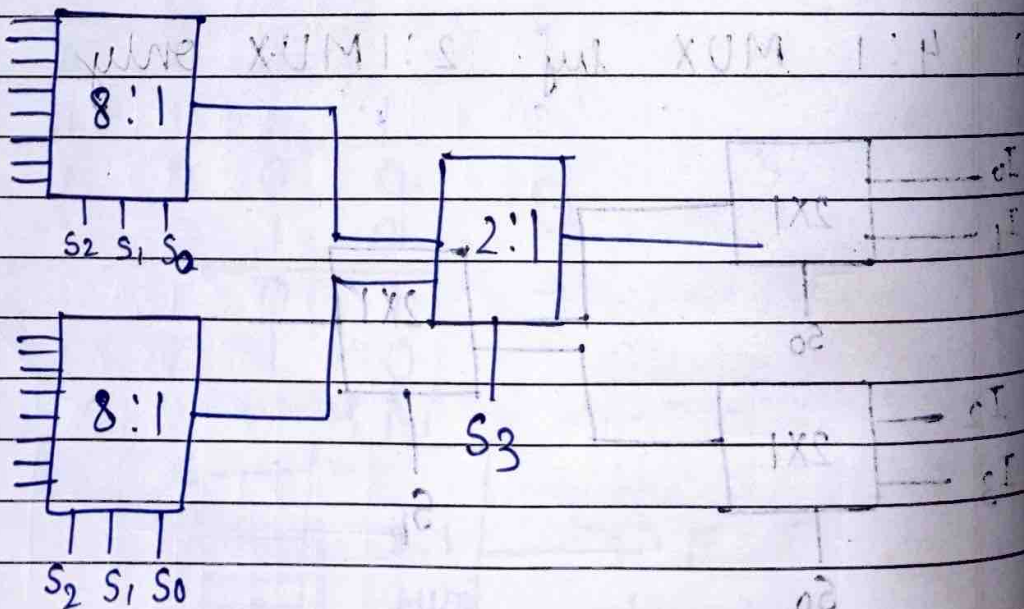
Q 4:1 MUX by 2:1 MUX only



Q 8:1 MUX by 2:1 MUX only



Q 16:1 MUX by 8:1 MUX



→ demultiplexer

- combinational circuit that receives single input signal to give multiple output signals
- also called as data distributor

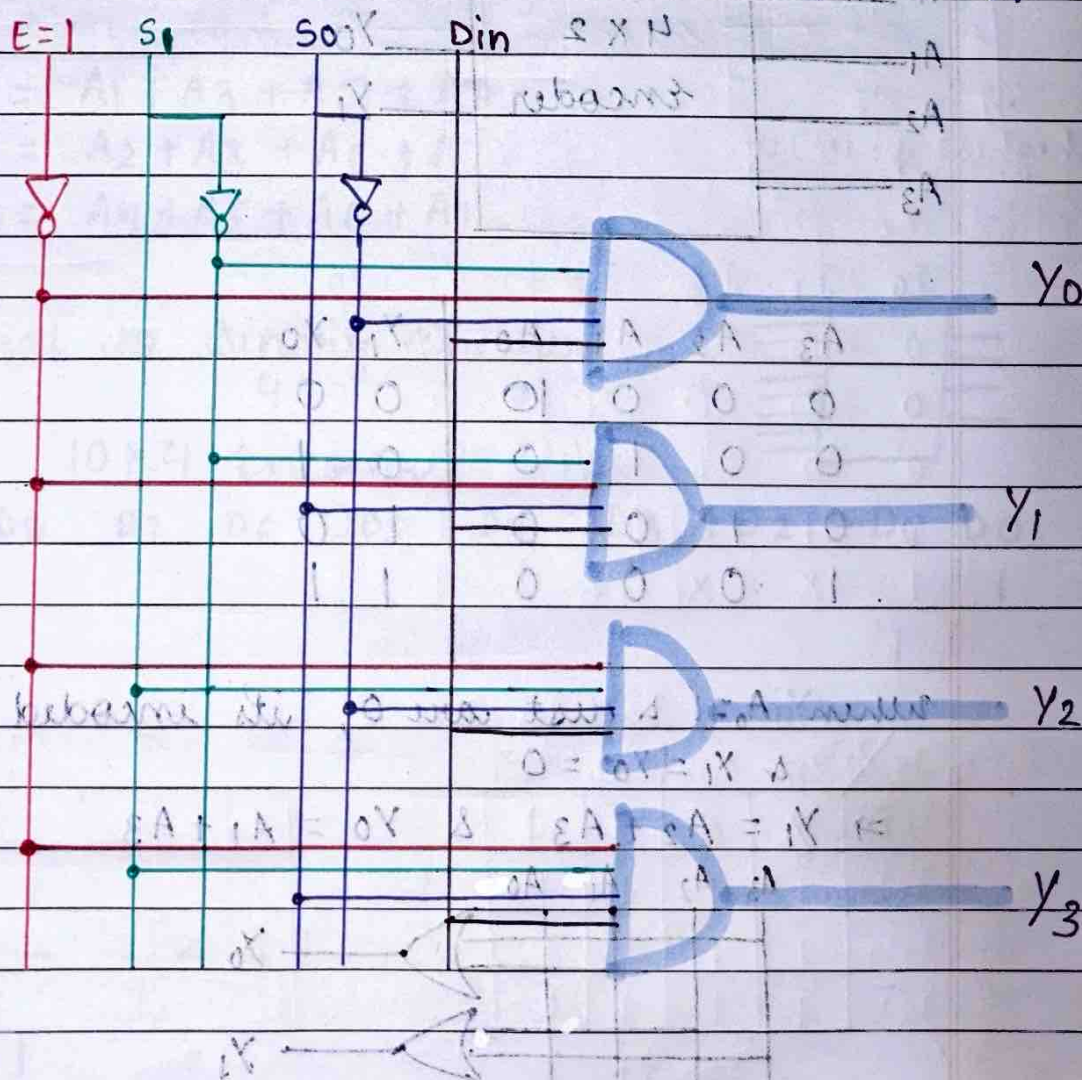
E	S ₁	S ₀	Y ₀	Y ₁	Y ₂	Y ₃
1	X	X	0	0	0	0
0	0	0	Din	0	0	0
0	0	1	0	Din	0	0
0	1	0	0	0	Din	0
0	1	1	0	0	0	Din

$$Y_0 = \bar{E} \bar{S}_1 \bar{S}_0 \text{ Din}$$

$$Y_1 = \bar{E} \bar{S}_1 S_0 \text{ Din}$$

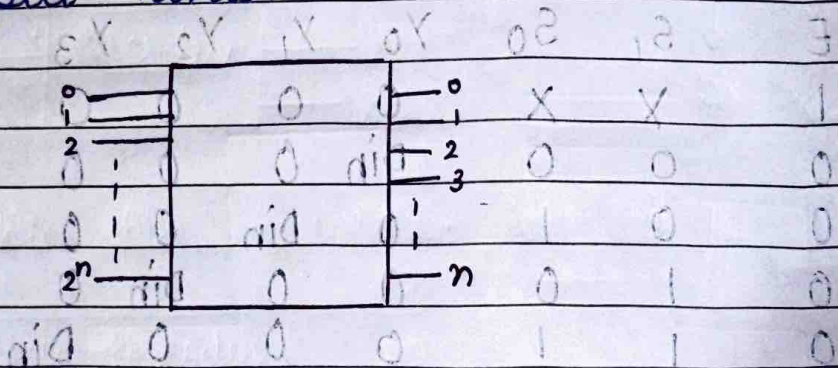
$$Y_2 = \bar{E} S_1 \bar{S}_0 \text{ Din}$$

$$Y_3 = \bar{E} S_1 S_0 \text{ Din}$$

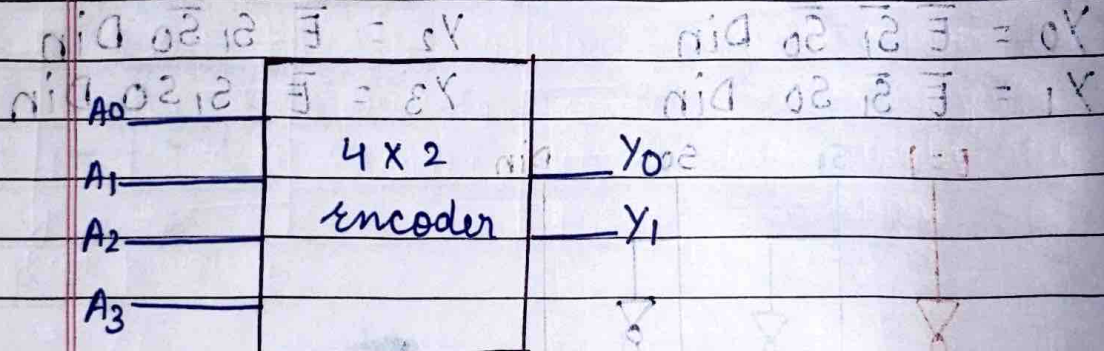


→ Encoder

- A combinational logic circuit that converts 2^n input lines into $\log_2 2^n$ bit binary output code.
- It has 2^n input lines to give n output lines.



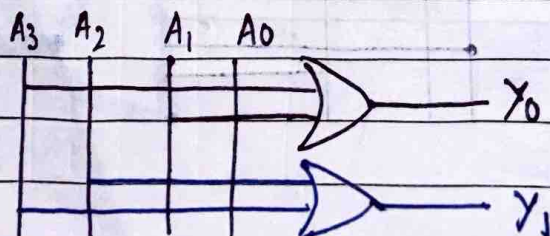
→ 4x2 encoder



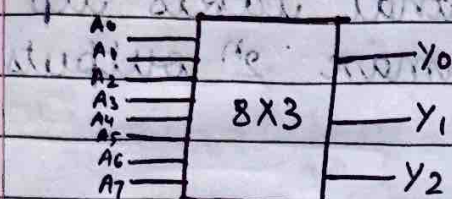
A_3	A_2	A_1	A_0	Y_1	Y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

When $A_0=1$ & rest are 0, it's encoded
& $Y_1=Y_0=0$

$$\Rightarrow Y_1 = A_2 + A_3 \quad \& \quad Y_0 = A_1 + A_0$$



→ Decimal to Binary (8X3) encoder



A7	A6	A5	A4	A3	A2	A1	A0	Y2	Y1	Y0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

$$Y_0 = A_1 + A_3 + A_5 + A_7$$

$$Y_1 = A_2 + A_3 + A_6 + A_7$$

$$Y_2 = A_4 + A_5 + A_6 + A_7$$

→ Decimal to Binary Encoder

10

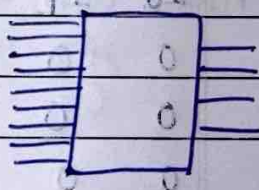
4

0

0

1

0



10X4 encoder

Y3 Y2 Y1 Y0 D9 D8 D7 D6 D5 D4 XD3 XD2 D1 D0

0 0 0 0 1 1 X X X 1 1

0 0 0 1 1 1

0 0 1 0 1 1

0 0 1 1 1 1

0 1 0 0 1 1

0 1 0 1 1 1

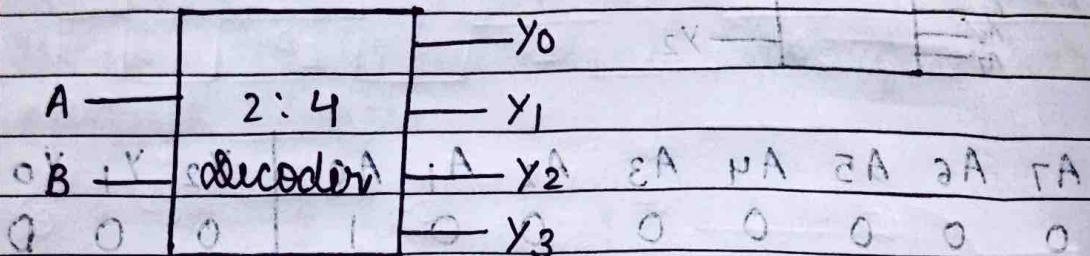
0 1 1 0 1 1

1 0 0 0 1 1

1 0 0 1 1 1

→ Decoder (2X8) provided at lab 10

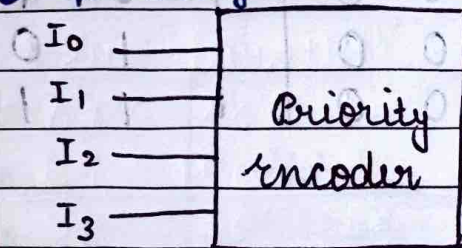
- combinational circuit that takes up n inputs to give max 2^n outputs



- enable pin must be set to 1

→ Priority encoder

lowest priority



again high priority

wali 1 hai 1 to low

wala se fir in pdta

high wala hi output

hoga = 0Y

$$Y_0 = I_0 + I_1 + I_2 + I_3$$

$$Y_1 = I_0 + I_1 + I_2 + I_3$$

$$Y_2 = I_0 + I_1 + I_2 + I_3$$

I₃ I₂ I₁ I₀ Y₁ Y₀

0 0 0 0 0 0

0 0 0 1 0 0

0 0 1 0 0 0

0 0 1 1 0 0

0 1 0 0 1 0

0 1 0 1 1 0

0 1 1 0 1 1

0 1 1 1 1 1

1 0 0 0 0 0

1 0 0 1 0 0

1 0 1 0 0 1

1 0 1 1 0 1

1 1 0 0 1 1

1 1 0 1 1 1

1 1 1 0 1 1

1 1 1 1 1 1

$$Y_1 = I_2 + I_3$$

$$Y_0 = I_3 + I_1 \bar{I}_2$$

I ₃	I ₂	I ₁ I ₀	I ₁ I ₀	I ₁ I ₀	I ₁ I ₀
I ₃ I ₂	X	0	0	0	0
I ₃ I ₂	1	1	1	1	1
I ₃ I ₂	1	1	1	1	1
I ₃ I ₂	1	1	1	1	1

For Y₁

X	0	1	1
0	0	0	0
X	X	X	X
1	1	1	1

For Y₂

x : ek ka
me 11 ka
the aur
me 0

→ Adder Unit consists of the following types

1. Half Adder

- Adds only single bit numbers
- Doesn't include carry from previous stage

A	B	Cout	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$S = A \oplus B$$

$$Cout = AB$$

2. Full Adder

- Adds multiple bit nos

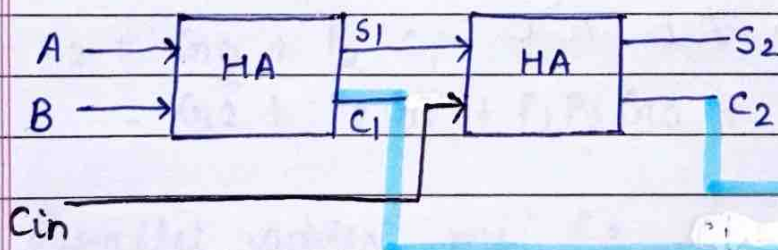
includes previous bit carry

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$S = A \oplus B \oplus Cin$$

$$Cout = AB + Cin(A \oplus B)$$

3. F.A. using H.A.



$$S_1 = A \oplus B$$

$$S_2 = A \oplus B \oplus Cin$$

$$C_1 = AB$$

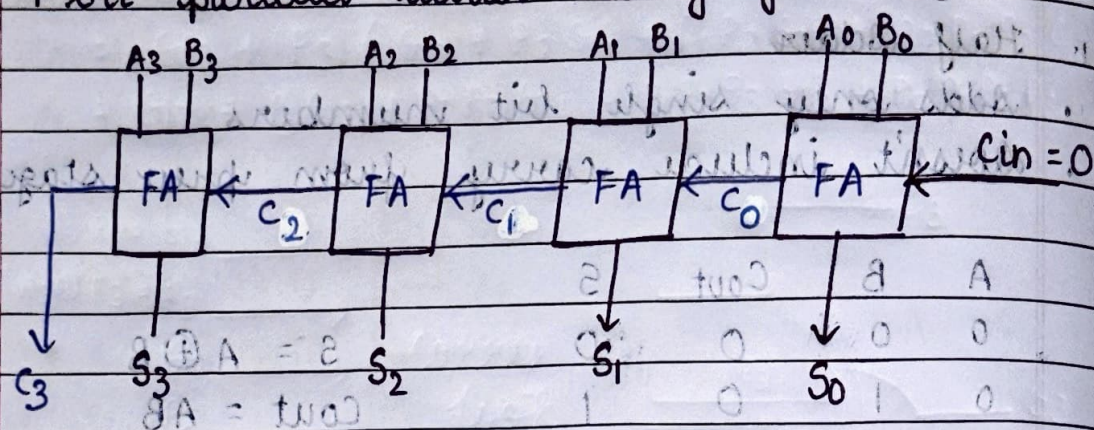
$$C_2 = S_1 Cin = (A \oplus B) Cin$$

$$Cout$$

$$= C_1 + C_2$$

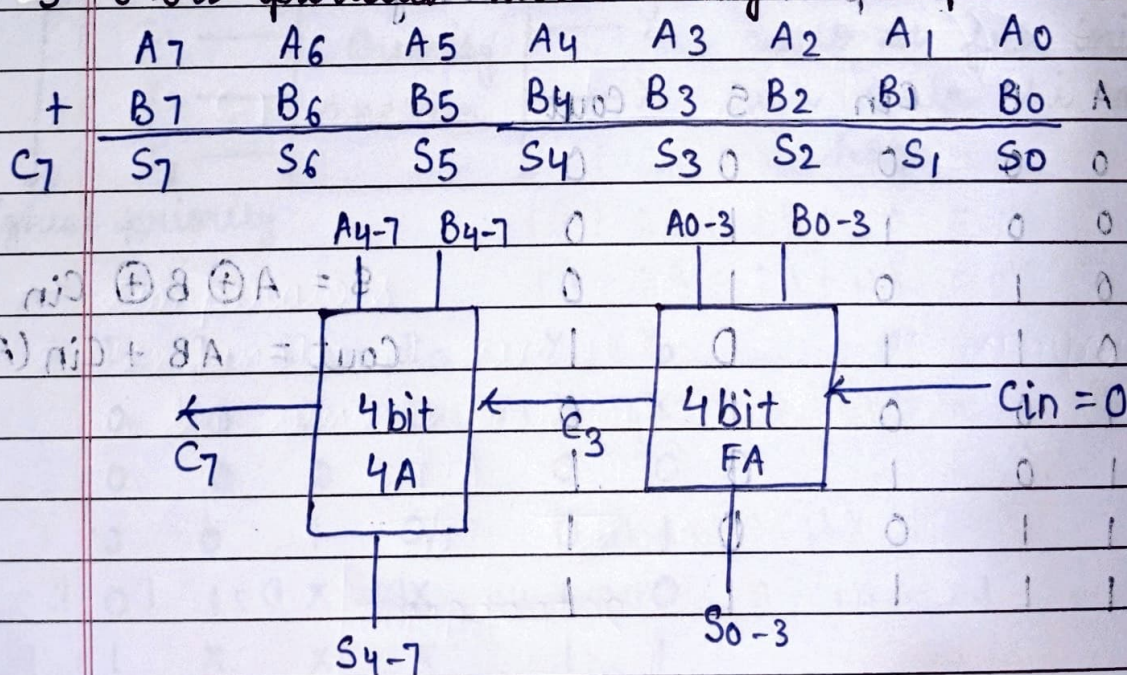
$$= AB + (A \oplus B) Cin$$

4. 4 bit parallel adder using full adder

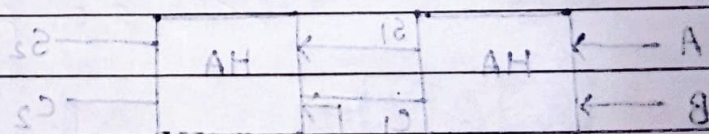


$$\begin{array}{r}
 A_3 \quad A_2 \quad A_1 \quad A_0 \quad 0 \\
 + B_3 \quad B_2 \quad B_1 \quad B_0 \\
 \hline
 C_3 \quad S_3 \quad S_2 \quad S_1 \quad S_0
 \end{array}$$

5. 8 bit parallel adder using 4 bit parallel adders



$$\begin{array}{r}
 A_7 \quad A_6 \quad A_5 \quad A_4 \quad A_3 \quad A_2 \quad A_1 \quad A_0 \\
 + B_7 \quad B_6 \quad B_5 \quad B_4 \quad B_3 \quad B_2 \quad B_1 \quad B_0 \\
 \hline
 C_7 \quad S_7 \quad S_6 \quad S_5 \quad S_4 \quad S_3 \quad S_2 \quad S_1 \quad S_0
 \end{array}$$



$$8A = 10$$

$$C_2 = 2^{10} = 1024$$

6. Carry look ahead adder

- (1.) Reduces the propagation delay (time ↓)
 (2.) CLA predicts the carry before it is calculated

A	B	Cin	Cout
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$Cout = AB + (A \oplus B) Cin$$

$$Cout = A \cdot B + (A \oplus B) Cin$$

$$Cout = AB + (A \oplus B) Cin$$

Carry generator (G)

Carry propagator (P)

But we know that $C_i = G_i + P_i C_{i-1}$

$$\Rightarrow C_i = G_i + P_i C_{i-1}$$

$$i=0 \quad C_0 = G_0 + P_0 C_{-1}$$

$$i=1 \quad C_1 = G_1 + P_1 C_0$$

$$C_1 = G_1 + P_1 (G_0 + P_0 C_{-1})$$

$$C_1 = G_1 + P_1 G_0 + P_1 P_0 C_{-1}$$

$$i=2 \quad C_2 = G_2 + P_2 C_1$$

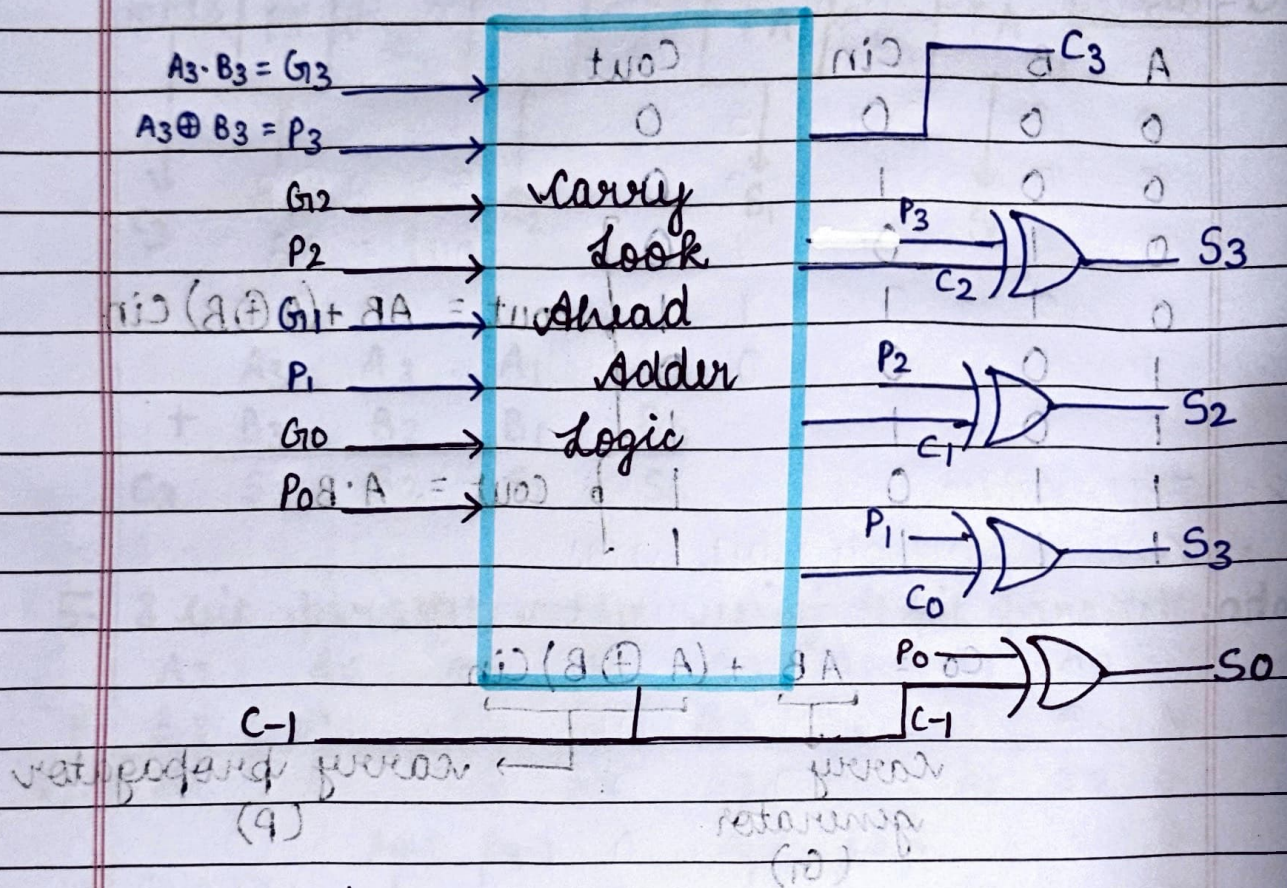
$$= G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_{-1}$$

Parallel adder me C_3 ki lie hume C_2, C_1, C_0 chahiye the but ab sirf C_{-1} (input carry) se hi carry nikal jayegi to time ↓ lagega

$$i=3 \quad C_3 = G_3 + P_3 C_2$$

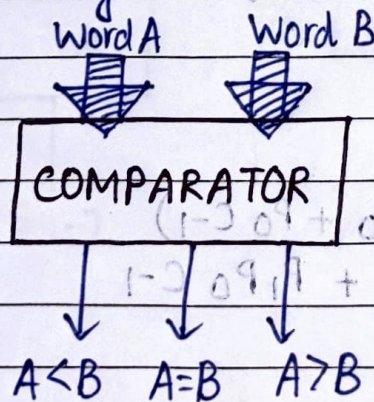
$$= G_3 + P_3 (G_2 + P_2 G_1 + P_1 P_2 G_0 + P_0 P_1 P_2 C_{-1})$$

$$\text{bitolunol} = G_3 + P_3 G_2 + P_2 P_3 G_1 + P_1 P_2 P_3 G_0 + P_0 P_1 P_2 P_3 C_{-1}$$



→ Comparators

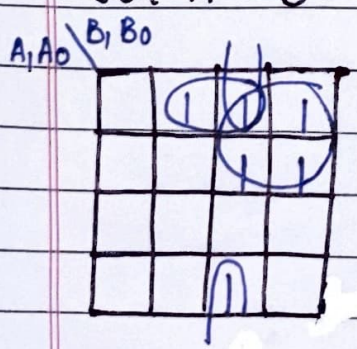
- Combinational circuit designed to compare two n -bit binary numbers.



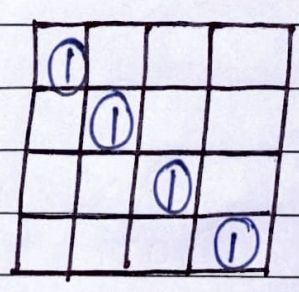
For 2 bit comparator

Inputs				Outputs		
A ₁	A ₀	B ₁	B ₀	A < B	A = B	A > B
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

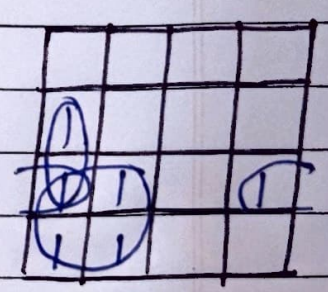
For A < B



For A = B



For A > B



$$(A < B) = \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_1 B_1 + \bar{A}_0 B_1 B_0$$

$$(A = B) = (A_1 \odot B_1) (A_0 \odot B_0)$$

$$(A > B) = A_0 \bar{B}_1 B_0 + A_1 A_0 \bar{B}_0 + A_1 \cdot \bar{B}_1$$